Worksheet: Deadweight Loss

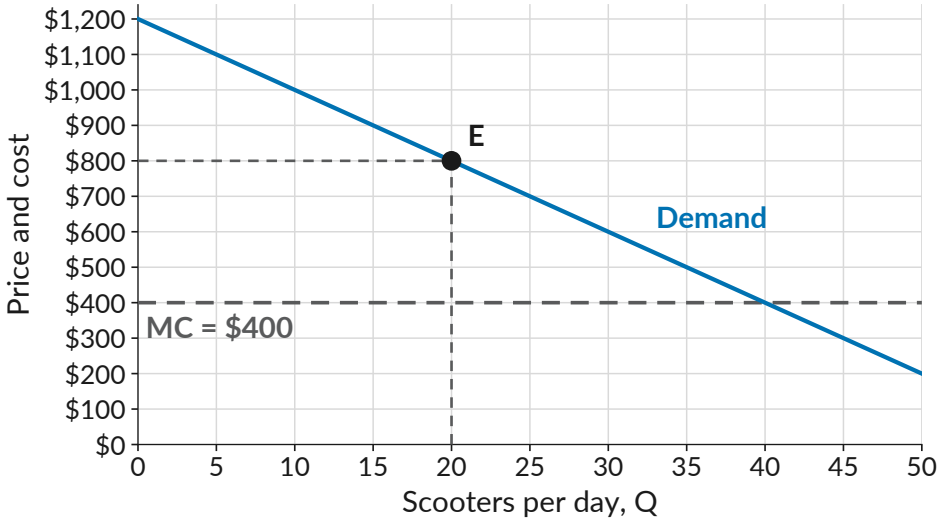
ECON 201

Definitions

- **Notation**: WTP is the buyer's willingness to pay, P is the price, and MC is marginal cost.
- **Gains from one sale**: the buyer gains WTP $-$ P and the firm gains P $-$ MC. The **joint surplus** is their sum, WTP $-$ MC.
- **Pareto improvement**: a change that makes at least one person better off without making anyone worse off.
- **Deadweight loss**: the surplus lost from sales that do not happen.
- **Areas**: a triangle is $\frac{1}{2} \times$ base $\times$ height, and a rectangle is base $\times$ height.

Electric scooters

A company sells electric scooters. The graph shows the demand curve for its scooters and its marginal cost. It maximizes profit at the point E, where it's $MR = MC$.



1. How many buyers are willing to pay more than the $400 it costs to make a scooter? How many scooters does the company sell?

2. How much is the 30th buyer willing to pay? Does this buyer buy at the company's price?

3. Suppose the company sold this buyer one scooter for $500, and kept its price at $800 for everyone else. What would each side gain? Is this a Pareto improvement?

4. Shade the deadweight loss on the graph and find its value.